

Day 12 Problem Set

Geometry Summer Credit | Special Right Triangles

Name:	Date:	Period:
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Goal

Use special right-triangle patterns fluently and keep exact radical answers.

Support Practice

1.	Simplify: $\sqrt{32}$.
2.	Simplify: $\sqrt{108}$.
3.	A 45-45-90 triangle has one leg 6. Find the other leg and hypotenuse.
4.	A 30-60-90 triangle has short leg 4. Find the long leg and hypotenuse.
5.	A 30-60-90 triangle has hypotenuse 14. Find the short leg.

Core Practice: 45-45-90

6.	A 45-45-90 triangle has leg 13. Find the hypotenuse.
7.	A square has side length 9. Find the diagonal.
8.	A 45-45-90 triangle has hypotenuse $5\sqrt{2}$. Find each leg.

9.	A square has diagonal $18\sqrt{2}$. Find the side length.
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10.	A 45-45-90 triangle has leg $4\sqrt{3}$. Find the hypotenuse.
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Continued

Core Practice: 30-60-90 and Mixed Review

11. A 30-60-90 triangle has short leg 7. Find the long leg and hypotenuse.

12. A 30-60-90 triangle has hypotenuse 30. Find both legs.

13. A 30-60-90 triangle has long leg $12\sqrt{3}$. Find the short leg and hypotenuse.

14. A 30-60-90 triangle has short leg $3\sqrt{2}$. Find the long leg and hypotenuse.

15. A ramp forms a 30-60-90 triangle. The short leg is 6 ft. Find the hypotenuse.

16. Choose the best tool and solve: right triangle with legs 10 and 10.

17. Choose the best tool and solve: right triangle with hypotenuse 20 and a 30 degree angle.

18. Spiral review: Find the distance between $(-3, 2)$ and $(5, 8)$.

19. Spiral review: Do 12, 16, and 20 form a right triangle?

20. Challenge: A student says a 30-60-90 triangle with hypotenuse 20 has long leg $10\sqrt{2}$. Correct the mistake.